



Importance of nanolayer formation in nanofluid properties: Equilibrium molecular dynamic simulations for Ag-water nanofluid

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ARTICLE INFO

Article history:

Received 4 January 2018

Received in revised form 4 March 2018

Accepted 25 May 2018

Available online 29 May 2018

Keywords:

Silver nanofluid

Liquid layering

Equilibrium molecular dynamics

Density

Viscosity

Ternary mixture

ABSTRACT

Most of the researches have authenticated the irrefutable role of interfacial nanolayer in molecular level. Understanding the mechanism of interfacial nanolayer effect on macroscopic fluid properties is of crucial importance. Thus, this study is aimed to examine the effect of nanolayer on two important nanofluid properties, i.e. density and viscosity, by performing extensive molecular dynamics simulations. It was found that, the formation of the ordered liquid nanolayer caused contraction of base fluid. The nanofluid was considered as a three-component mixture, i.e. nanoparticle, base fluid after mixing with nanoparticles and nanolayer. Accordingly, a new ternary mixture relation was proposed to compute the nanofluid density. Obtained results revealed the importance of nanolayer effect on density of nanofluids. Moreover, the role of liquid layering in nanofluid viscosity was analyzed. The obtained results can provide useful understanding on the structural and transport properties of nanofluids not only in the Ag-water nano-colloid, but also in other types of nanofluids.

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1. Introduction

Recognition of nanoscale mechanisms is crucial to provide a better explanation for nanocolloids properties. The interfacial nanolayer is one of the most important nanoscale mechanisms playing a significant role in thermophysical properties of nanofluids. Coulomb or van der Waals interactions at solid-liquid interface tend to form a dense liquid layer which has higher order around the nanoparticle. Numerous studies have attempted to model the thermophysical properties of nanofluid considering the role of this ordered nanolayer.

Keblinski et al. [1] put forward four specific mechanisms that could have key role in thermal conductivity of nanofluid. Liquid layering at the liquid-particle interface was one of the studied mechanisms.

Yu and Choi considered the role of solid-liquid interfacial layer and developed a renovated Maxwell model [2] which was limited to suspensions with spherical particles and a renovated Hamilton–Crosser model [3] for suspensions consisting of non-spherical particles. Although these proposed models showed improved increase in nanofluid thermal conductivity, they failed to predict the nonlinear behavior of thermal conductivity augmentation.

Other researchers have also presented models that take into account the effect of nanolayer in thermal conductivity of

nanofluids [4–9]. Avsec and Oblak [9] presented mathematical models for computing the thermophysical properties of nanofluids. They considered an elliptical liquid layer shell around nanoparticles in extracting models to calculate the thermal conductivity and viscosity of nanofluids. Their analytical results were compared with the experimental data and showed relatively good agreement.

Yang et al. [10] proposed two models for calculating the viscosity of ammonia water nanofluids by considering the solvation effect. Both of their models were based on Einstein model. One of their models analyzed a monolayer adsorption without considering the decrease of surfactant content in ammonia water-based fluid. The other one analyzed the electric double layer (EDL) adsorption, by considering the change of surfactant content in ammonia water-based fluid. Their results revealed that both models could predict the viscosity of ammonia water nanofluid in better agreement with the experimental data.

The effect of nanolayer on nanofluid density has been also considered in a few studies. In this regard, Sharifpur et al. [11] investigated the nanofluid density experimentally and showed that the mixture model for density of nanofluids had higher values than the obtained experimental data. They interpreted that the reason for such difference could be attributed to the interfacial layer between particles and base fluid. They developed a new model for predicting the nanofluid density which included the role of nanolayer.

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Nomenclature

d	particle diameter (nm)
E	energy (J)
F	force (N)
m	mass (kg)
M	molecular mass (gr/mol)
p	momentum (kg·m/s)
q	partial charge ratio
r	distance (m)
t	nanolayer thickness (nm)
T	temperature (K)
U	potential energy (J)
V	volume (m ³)

Subscript

bf	base fluid
f	final state, after mixing
i	initial state, before mixing
nf	nanofluid
nl	liquid nanolayer
np	nanoparticle

Greek letters

φ	volume fraction
ε	potential well depth
σ	cut-off distance
ρ	density (kg/m ³)
μ	shear viscosity (Pa·s)
τ_{xy}	shear stress (Pa)

Constants

K_B	Boltzmann constant
K_C	electrostatic constant
N_A	Avogadro constant

Molecular dynamic method is a powerful computational tool for simulating materials in molecular scale [12]. It is a suitable and reasonable method capable to evaluate the thermophysical properties while being applied in recognition of molecular phenomena [13–18]. Several researches have been devoted to simulation and evaluation of the molecular layering at liquid-solid interface.

Li et al. [19] assessed the interfacial molecular layering in a copper nanofluid by equilibrium molecular dynamics simulation. They analyzed the density distribution of the liquid near the nanoparticle and identified that the thickness of the layering was about 0.5 nm under the factors used in their study.

Equilibrium thermodynamic properties of gold-water nanolayer mixtures in various volume fractions were investigated by Puliti et al. [20] using molecular dynamic simulations. They used accurate force-fields for precise modeling of inter- and intra-molecular interaction of the components. They verified their results using pure fluid and solid properties. They reported non-ideal behavior of the system due to several factors including liquid layering, anisotropy and other solid-liquid interfacial effects.

Cui et al. [21] used molecular dynamics simulation to study the liquid-solid interfacial absorption layer in argon-based nanofluids. They considered different influential parameters such as nanoparticle shape, size and material. By comparing the number density distributions of nanofluids, they concluded that the absorption layer

included two denser locations of liquid atoms whereas the first denser location was influenced by the bulk material attribute and the size of nanoparticle, while the second denser location was mainly affected by the evolution time. Moreover, the microscopic structure of the absorption layer was more ordered and more similar to that of solid material.

On the basis of our literature review, so far, a few works have addressed the effect of nanolayer on two important thermophysical properties of nanofluids, namely density and viscosity. Therefore, this paper was aimed to investigate the role of nanolayer formation in determining the density and viscosity of nanofluids. In this regard, molecular dynamic simulations of Ag-water nanofluids were carried out. At first, the structure of absorption nanolayer was analyzed and then the effect of this layer on density and viscosity of nanofluid were examined. Furthermore, a new correlation based on ternary mixture for calculating the nanofluid density was proposed.

2. Simulation method and models

Water has been modeled using computationally efficient model of TIP4P/2005 [22]. The TIP4P/2005 model is the best accurate four-site force-field that simulate structure of water, equation of state, surface tension and water transport properties such as thermal conductivity and viscosity in good agreement to experimental data [23,24]. This model specifies four-site rigid water molecules with charges. It ameliorates the electro-static distribution around water molecules. Electrostatic interactions and distribution forces are also modeled using Coulombic model and Lenard-Jones (LJ) potential, respectively. The corresponding potential between two water molecules “a” and “b” is calculated by Eq. (1):

$$E_{ab} = \sum_i \sum_j \frac{K_c q_i q_j}{r_{ij}} + U_{LJ} \quad (1)$$

where K_c is electro-static constant, q_i is the partial charge ratio (to electron charge) and r_{ij} is the distance between two atoms. Charged points are located on atoms or artificial (dummy) points. The long-term interactions which are simulated by Coulombic model use particle-particle-mesh (pppm) method. To model the inter-atomic interactions, pair-wise LJ potential is utilized which is calculated using Eq. (2):

$$U_{LJ} = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] \quad (2)$$

where σ and ε are potential well depth and cut-off distance, respectively, and r is inter-atomic distance. This potential has lower computational costs compared to other potentials for modeling the particles.

To smooth the energy and force in Coulombic model and LJ potentials, a switching function of $S(r)$ is also applied as:

$$S(r) = \frac{[r_{c,out}^2 - r^2]^2 [r_{c,out}^2 + 2r^2 - 3r_{c,in}^2]}{[r_{c,out}^2 - r_{c,in}^2]^3} \quad (3)$$

where $r_{c,in}$ and $r_{c,out}$ are inner and outer cut-off radii which values are 8 and 10, respectively.

Silver nanoparticles with 4 nm diameter (1985 Ag atoms) have been considered in water as a base fluid. Water molecules have encompassed nanoparticle and randomly distributed around it. In this study, a simulation box including 26,892 water molecules with dimensions of 10 nm × 10 nm × 10 nm is considered. Periodic boundary conditions are applied in all three spatial directions. In order to prevent Brownian motion of the nanoparticle in the water, a spring with constant of 100 kcal/molÅ² connects center of the nanoparticle to a virtual point at the center of the simulation box.

Interatomic forces in silver nanoparticle are modeled using LJ potential taken from [25]. Water-Ag interactions are calculated regarding to combination rules of Lorentz-Berthelot:

$$\sigma_{ij} = \frac{\sigma_{ii} + \sigma_{jj}}{2} \quad (4)$$

$$\epsilon_{ij} = \sqrt{\epsilon_{ii}\epsilon_{jj}} \quad (5)$$

where i and j signify non-bonding atoms in the system. Table 1 presents the LJ parameters, i.e., ϵ and σ for different atomic interactions.

Geometry of a silver nanoparticle confined in the base fluid of water is demonstrated in Fig. 1.

Molecular dynamic simulation was firstly implemented in NPT ensemble in order to reach to desired pressure and temperature with time step of 1fs. Then the system was run in NVT ensemble to equilibrate. After equilibrium conditions, four million time steps were performed and the results has been averaged during the steps. All MD simulations were performed using Large-scale Atomic-molecular Massively Parallel Simulator Package (LAMMPS) [27].

2.1. Water density and validation

In order to validate our MD method, the density of pure water was calculated at different temperatures. In all temperatures, the pressure of the system was set to 1 bar. Fig. 2 presents MD results and comparison with experimental data for density of pure water in the range of 25 to 65 °C. As it is shown, there are good agreements between MD results and experimental data for the density of pure water which present the good accuracy of water model in this simulation.

2.2. Water viscosity and validation

Viscosity of system is modeled by Green-Kubo relation which is on the basis of shear stress (τ_{xy}) fluctuations:

$$\mu = \frac{V}{k_B T} \int_0^\infty \langle \tau_{xy}(t) \tau_{xy}(0) \rangle dt \quad (6)$$

where

$$\tau_{xy}(t) = \sum_i \frac{p_x^i p_y^i}{m^i} - \frac{1}{2} \sum_i r_x^{ij} F_y^{ij} \quad (7)$$

V and T are volume and temperature of the system, respectively, and k_B is Boltzmann constant. m is mass and p is momentum as well as r , F are distance and force between two particles, respectively.

In order to validate our MD method, the viscosity of pure water was calculated at different temperatures. In all temperatures, the pressure of the system was set to 1 bar. Fig. 3 presents MD results and comparison with experimental data for viscosity of pure water in the range of 25 to 65 °C. As it is shown, there are good agreements between MD results and experimental data for the viscosity of pure water which present the good accuracy of water model in this simulation.

Table 1
LJ parameters for non-bonding interactions [26].

Interaction type	ϵ (kcal/mol)	σ (Å)
Ag-Ag	7.95	2.64
H ₂ O-H ₂ O	0.182	3.16
Ag-H ₂ O	1.21	2.9

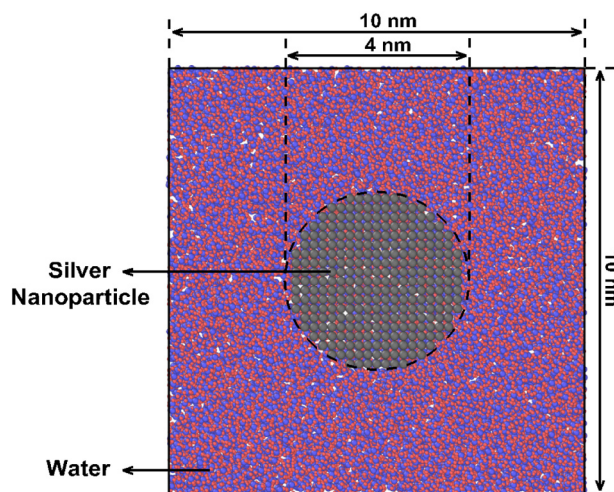


Fig. 1. Silver nanoparticle in base fluid of water, the boundary conditions are periodic in all spatial directions.

3. Results and discussion

3.1. Nanolayer thickness

Through the investigation of the density of water molecules around the silver nanoparticle, it is understood that the density of water molecules is not uniform in the computational domain. The density profiles of surrounding water in different nanoparticle diameter are illustrated in Fig. 4. As it can be seen, the particle affected the density so that it is not monotonically distributed around the particle. Density is higher near the particle and it gradually decreases while getting away, till density of pure water is achieved. The density profile is almost the same for all nanoparticles with different diameters. However, there is a fast decay over a distance of 0.3 nm from the particle surface which can be interpreted as an intense influence of the particle attendance on the density of vicinity water. It is clear that beyond a distance of 1 nm the density of pure water is roughly achieved. Therefore, a thickness between 0.3 nm and 1 nm can be assumed to be the thickness of the ordered nanolayer, which is consistent with the results of previous researches [19,21]. Moreover, the thickness of nanolayer can be

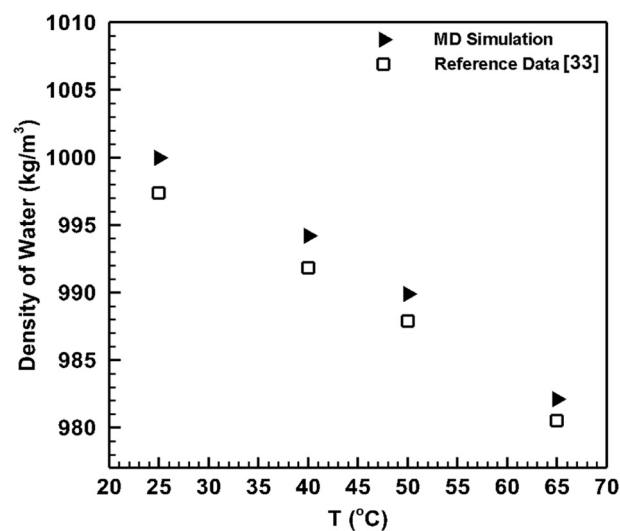


Fig. 2. Comparison of MD results and Reference data for density of pure water at various temperatures.

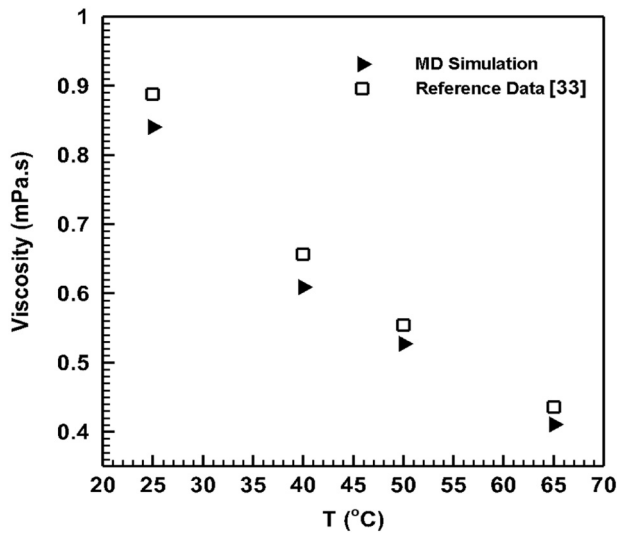


Fig. 3. Comparison of MD results and experimental data for viscosity of pure water at various temperatures.

estimated from the Langmuir formula of monolayer adsorption of molecules as [28]:

$$t = \frac{1}{\sqrt{3}} \left(\frac{4M_{bf}}{\rho_{bf}N_A} \right)^{1/3} \quad (8)$$

where M_{bf} is the molecular weight of liquid, ρ_{bf} is the density of liquid, and N_A is Avogadro constant ($6.023 \times 10^{23} \text{ mol}^{-1}$). In our system, with $M = 18.02 \text{ g/mol}$ and $\rho_{bf} = 1 \text{ g/cm}^3$, the estimated adsorption thickness will be around 0.28 nm which is in good agreement with MD result. Density contour of surrounding water is also depicted in Fig.5.

The average nanolayer density can be obtained through volumetric integration on the results of MD simulation up to the prescribed thickness of nanolayer. Since in practice the results of MD simulation for the large magnitude of nanoparticle diameter are not available,

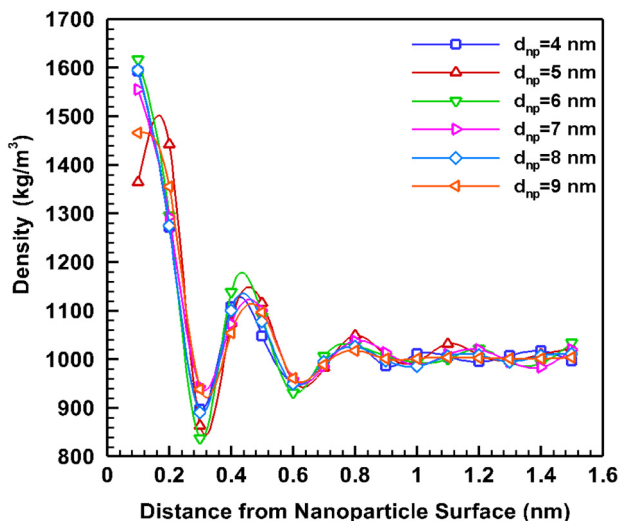


Fig. 4. Density change of water near the nanoparticle for different nanoparticle diameter.

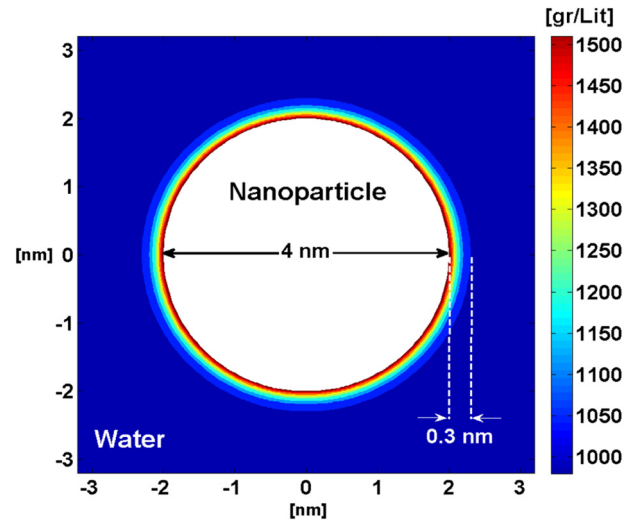


Fig. 5. Contour of density distribution of water molecules around the nanoparticle.

therefore, it is necessary to recommend a relation that derives the density of nanolayer for every diameter from a curve fitting on existing MD data.

Similar to Ref. [29] for thermal conductivity of nanolayer, a density profile is assumed by considering that in limiting conditions, i.e. $d_{np} \rightarrow 0$ and $d_{np} \rightarrow \infty$, the density of nanolayer should be vanished and approached to a constant value, respectively.

$$\rho_{nl}(d_{np}) = ae^{-\frac{b}{d_{np}^c}} \quad (9)$$

The best curve fit coefficients for different nanolayer thicknesses are listed in Table 2. The proposed Eq. (9) for different nanoparticle diameters and nanolayer thicknesses is depicted in Fig. 6. The conformity of MD data with proposed equation is evident. The proposed Eq. (9) will be applied in calculations of the next sections.

3.2. Density of nanofluid

One of the important physical properties of the nanofluids is density. This property has a key role in flow and heat transfer behavior of the fluid and directly affects the Reynolds number, pressure loss, and Nusselt number. For example, the natural convection is caused by density variations due to temperature differences. This property also has an important role in determining some of the properties such as the thermal diffusivity.

The classical formula which has been often used in calculating the density of nanofluids is based on solid-liquid mixture model [30]. This formula is expressed as

$$\rho_{nf-old} = \rho_p\phi + \rho_{bf}(1-\phi) \quad (10)$$

where ρ_{nf} is the density of the nanofluid, ρ_{bf} is the density of the base fluid, ρ_p is the density of the nanoparticle, and ϕ is the nanoparticle

Table 2

The curve fitting coefficients for Eq. (9) in predicting the density of nanolayer.

Nanolayer thickness, t (nm)	a (kg/m ³)	c	b (nm)
0.3	1400	2.01	0.5
0.5	1188	1.76	0.5
1.0	1079	1.71	0.5

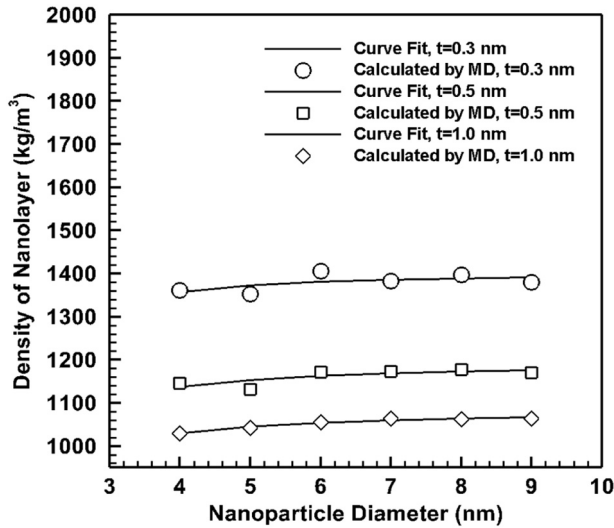


Fig. 6. MD simulation results and proposed curve fit (Eq. (9)) for the density of nanolayer.

volume concentration. The nanofluid volume fraction in the above formula is calculated from Eq. (11);

$$\varphi = \frac{V_{np}}{V_{nf}} = \frac{V_{np}}{V_{np} + V_{bf,i}} \quad (11)$$

where $V_{bf,i}$ is the volume of base fluid before mixing with nanoparticles.

Many researchers have used Eqs. (10) and (11) to calculate the density of nanofluid [31]; however, there are several experimental studies which reported remarkable deviation between their experimental data and calculated values from Eq. (10) (for example see Refs [11, 32]).

By dispersing the nanoparticles in the base fluid, an ordered dense nanolayer forms around nanoparticles. Therefore, the formation of these layers within the fluid can lead to reduction in total initial volume of the fluid. To develop a model to include the nanolayer effect on nanofluid density, mass balance of water before and after mixing with nanoparticles is considered to explain how the volume of water changes as below;

$$\rho_{bf} V_{bf,i} = \rho_{NL} V_{nl} + \rho_{bf} V_{bf,f} \quad (12)$$

$$V_{bf,f} = V_{bf,i} - \frac{\rho_{NL}}{\rho_{bf}} V_{nl} \quad (13)$$

where V_{nl} is the nanolayer volume which can be obtained from Eq. 14:

$$V_{nl} = \frac{4}{3} \pi \left(\left(\frac{d_{np}}{2} + t \right)^3 - \left(\frac{d_{np}}{2} \right)^3 \right) \quad (14)$$

where t is the thickness of nanolayer. Thus, the total volume of nanofluid can be written as;

$$V_{nf,new} = V_{np} + V_{bf,i} + V_{nl} \left(1 - \frac{\rho_{nl}}{\rho_{bf}} \right) \quad (15)$$

A new relation for volume fraction of nanofluid can be found when nanolayer is involved as;

$$\varphi_{new} = \frac{V_{np}}{V_{nf,new}} = \frac{V_{np}}{V_{np} + V_{bf,i} + V_{nl} \left(1 - \frac{\rho_{nl}}{\rho_{bf}} \right)} \quad (16)$$

Since the density of the nanolayer is higher than the density of water, $V_{nf,new}$ is always little than V_{nf} . Therefore, the new volume

fraction of nanofluid, φ_{new} , is greater than the old volume fraction, φ , obtained from Eq. (11). It should be noted that in Eq. (16), V_{nl} and ρ_{nl} are calculated from Eqs. (14) and (9), respectively.

It is possible to introduce a new binary mixture relation for calculating the nanofluid density based on the new volume fraction of nanofluid, namely;

$$\rho_{nf-new} = \rho_{np} \varphi_{new} + \rho_{bf} (1 - \varphi_{new}) \quad (17)$$

Fig. 7 depicted the density values of nanofluid obtained from MD simulation, new and traditional binary mixture models. All calculations are performed based on the old volume fraction of 0.04. The density of nanofluid from MD simulation can be computed directly from Eq. (18);

$$\rho_{nf} = \frac{(m_{water} + m_{Ag})}{V_{nf,MD \ box}} \quad (18)$$

where m_{water} is the total mass of water molecules, m_{Ag} is the total mass of silver atoms, and $V_{nf,MD \ box}$ is the equilibrium volume of the MD simulation box.

As can be seen, the results of the new binary mixture model are closer to the MD simulation results compared to the traditional binary mixture; however, the difference is still high.

One of the main assumptions of mixing theory is that when the materials are mixed together, the nature of the initial material does not change. By mixing the nanoparticles and base fluid a new solid-like ordered material called nanolayer is formed and so there will be a ternary mixture. Based on the ternary mixture approach the density of nanofluid can be written as;

$$\rho_{nf} = \varphi_{np,new} \rho_{np} + \varphi_{nl,new} \rho_{nl} + (1 - \varphi_{nl,new} - \varphi_{np,new}) \rho_{bf} \quad (19)$$

where

$$\varphi_{np,new} = \frac{V_{np}}{V_{nf,new}} \quad (20)$$

$$\varphi_{nl,new} = \frac{V_{nl}}{V_{nf,new}} \quad (21)$$

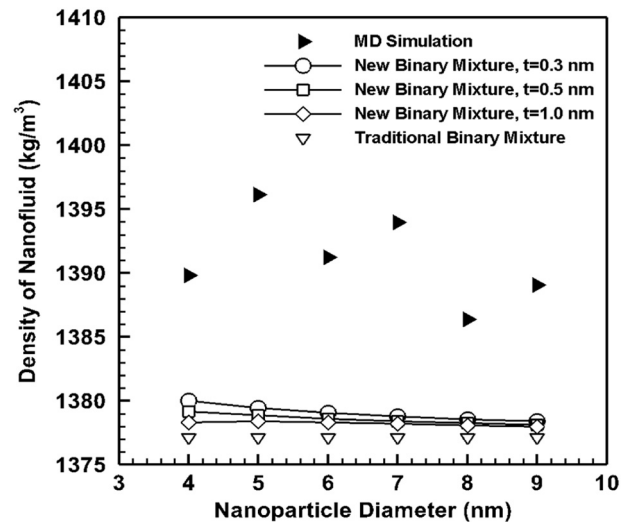


Fig. 7. Comparison of results between MD simulations, new and traditional binary mixture models, Eqs. (18), (17) and (10), respectively.

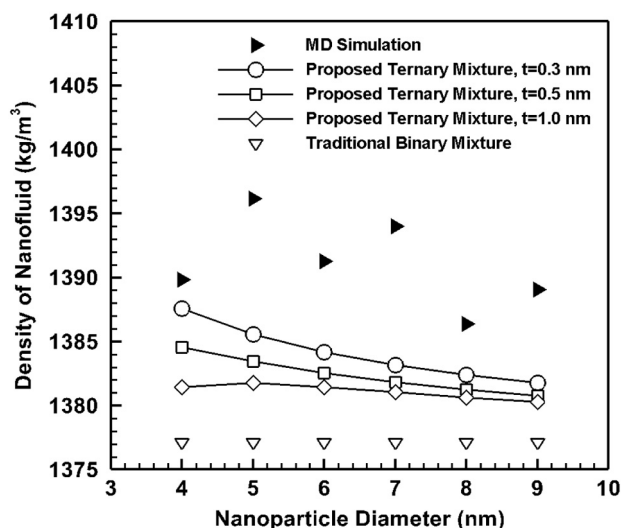


Fig. 8. Comparison of results between MD simulations, traditional binary mixture and proposed ternary mixture, Eqs. (18), (10) and (19), respectively.

$V_{nl,new}$ and $V_{nf,new}$ are computed from Eqs. (14) and (15), respectively. Moreover, the total volume of nanoparticles can be obtained as

$$V_{np} = \frac{4}{3} \pi \left(\frac{d_{np}}{2} \right)^3 \quad (22)$$

The results of proposed relation for computing the nanofluid density, i.e. Eq. (19), are depicted in Fig. 8. It is clear that the results are much closer to the MD simulation ones compared to what presented in Fig. 7. However, there are still little differences between the MD and proposed ternary mixture data.

So far, the bulk values of the properties of water and silver [34] have been used in calculations. Since the bulk values of materials are available, the use of them in calculations is appropriate. However, it is better to compare the results of proposed ternary mixture with the MD data. Thus, the MD simulation values for properties of water and silver are applied. The outcomes of this approach are presented in Fig. 9. Interestingly, there is an excellent agreement between the proposed ternary mixture data and MD simulation results while the MD simulation values for properties of water and silver are applied. Thus, the proposed

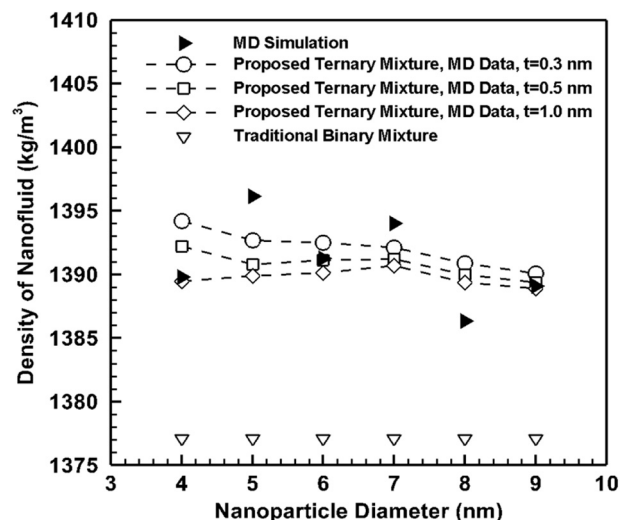


Fig. 9. Comparison of results between MD simulations Eq. (18), traditional binary mixture Eq. (10), and proposed ternary mixture Eq. (19), based on MD data for properties of water and silver.

ternary mixture model can be an inspiring way to compute the nanofluid density.

3.3. Viscosity of nanofluid

One of the important transport properties of nanofluids is viscosity. The viscosity is a measure of the resistance of a fluid to deformation under shear stress. Viscosity describes a fluid's internal resistance to flow and may be thought of as a measure of fluid friction. The most important role of this property is appeared in the frictional pressure drop of the system through which the fluid is flowing, as fluid with the higher viscosity has a higher pressure drop.

There are some existing models for estimating the viscosity of nanofluids. The pioneer model to predict the nanofluid viscosity is the well-known Einstein's model for a viscous fluid containing a dilute suspension of small, rigid, spherical particles [35]. This model is stated as;

$$\mu_{nf} = \mu_{bf}(1 + 2.5\varphi) \quad (23)$$

This formula has been found valid for relatively low particle loading ($\varphi \leq 2\%$).

Brinkman [35] has modified Einstein's equation to a more generalized formula for use with moderate particle concentrations as follows:

$$\mu_{nf} = \frac{\mu_{bf}}{(1 - \varphi)^{2.5}} \quad (24)$$

Considering the effect of Brownian motion of particles, Batchelor [35] proposed the following formula for an isotropic suspension of spherical and rigid particles:

$$\mu_{nf} = \mu_{bf}(1 + 2.5\varphi + 6.5\varphi^2) \quad (25)$$

There are few models that include the nanolayer effect in viscosity of nanofluids. Avsec and Oblak [9] considered an elliptical liquid layer shell around nanoparticles in extracting models for calculating the viscosity of nanofluids. Their model is also known as renewed Ward model and can be written as

$$\mu_{nf} = \mu_{bf} \left[1 + 2.5\varphi_{eff} + (2.5\varphi_{eff})^2 + \dots \right] \quad (26)$$

$$\varphi_{eff} = \varphi \left(1 + \frac{t}{r} \right)^3 \quad (27)$$

where, φ is the volume fraction, r and t are nanoparticle radius and liquid layer thickness, respectively.

Yang et al. [10] proposed two models for calculating the viscosity of ammonia water nanofluids by considering the solvation effect. Both of their two models were based on Einstein model. One of their proposed models assumed a monolayer adsorption without considering the decrease of surfactant content in ammonia water base fluid. The other model was to analyze electric double layer (EDL) adsorption, while taking into account the change of surfactant content in ammonia water base fluid.

Fig. 10 compares the results of the relative dynamic viscosity, i.e. the ratio of the nanofluid viscosity to the base fluid viscosity, calculated by MD simulations and those from various theoretical models in different volume fractions. Nanofluid has higher viscosity than pure water in all volume fractions and the relative viscosity is higher for higher volume fractions. Obviously, the results of models which don't include the nanolayer effect are not compatible with MD simulations and underestimate the viscosity in whole volume fractions. On the other hand, Avsec and Oblak [9] model and Yang et al. [10] model which considered the liquid nanolayer shell around nanoparticles follow the MD simulation

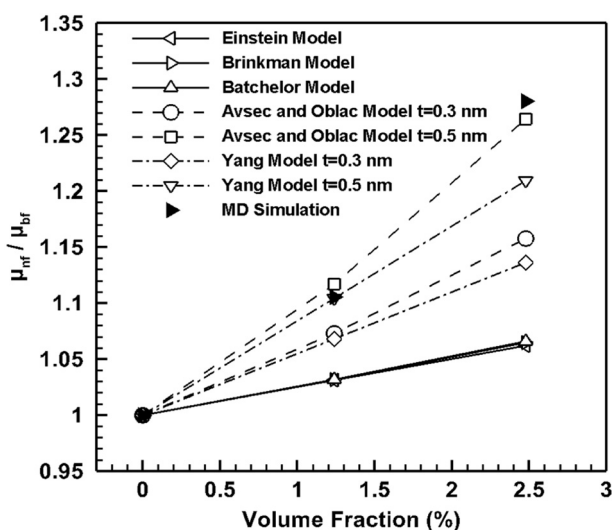


Fig. 10. Comparison of nanofluid viscosity between MD simulation and previous models.

results well. Hence, it can be concluded that the nanolayer can play an important role in viscosity of nanofluids.

4. Conclusion

In this research, equilibrium molecular dynamic simulations were used to investigate the role of nanolayer on two main thermophysical properties of nanofluid, i.e. density and viscosity. Obtained results revealed that the role of nanolayer as a key nanoscale mechanism is undeniable. In fact, the formation of the ordered liquid nanolayer causes the base fluid to contract, which will reduce the volume of the base fluid. Thus, the traditional volume fractions should be redefined based on new volume concentrations of three components, i.e. nanoparticle, base fluid after mixing with nanoparticles and nanolayer. Accordingly, a new ternary mixture relation was proposed to compute the nanofluid density.

The consequences of this research contribute to a better recognition of nanofluid behaviors in order to improve its applications in industry. Nevertheless, many additional researches are required to develop our knowledge toward nanofluids.

Acknowledgement

Iran National Science Foundation (INSF) is gratefully acknowledged for financial support of this work (Project No.92003035).

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